

Limited supervision, contamination, and topology transfer in collider anomaly detection: A reproducible LHC Olympics benchmark

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We study unsupervised and semi-supervised anomaly detection on the simulated LHC Olympics 2020 research and development dataset. A shared event split and five standardized jet observables support a controlled comparison of autoencoders, Isolation Forest, Deep SVDD, and a Deep SAD implementation, with fully supervised and mass-aware references. The released core comprises 23 configurations at one training seed and six reference configurations repeated at four additional seeds, for 47 independent fits. At the reference settings, the mean nominal-signal area under the receiver operating characteristic curve is 0.8068 ± 0.0077 for Isolation Forest and 0.8880 ± 0.0114 for Deep SAD with 100 known anomalies and 0.5% injected training contamination; uncertainties here describe training-seed dispersion. Paired event bootstrapping separately quantifies finite-test uncertainty. Transfer to a three-prong signal reduces the mean Deep SAD area to 0.7492 ± 0.0614 , while the fully supervised reference reaches 0.9292 ± 0.0012 . Mass-distribution diagnostics and an explicitly approximate sideband-fit study expose the distinction between discrimination and trustworthy discovery inference. Archived particle-level and density-calibration extensions provide exploratory checks rather than equivalent reproductions of published state-of-the-art methods. The contribution is an auditable empirical benchmark and its limitations, rather than a new learning objective or a calibrated discovery claim.

I. INTRODUCTION

Searches for new particles often specify a signal model before defining discriminating observables. Anomaly detection instead attempts to identify departures from a reference distribution, potentially retaining sensitivity to signals outside a narrowly chosen training hypothesis. The LHC Olympics challenge established a public setting for assessing such approaches and demonstrated the variety of available supervision strategies [1]. This motivation does not remove the need for an explicit reference distribution, an evaluation protocol, or a statistical model for a claimed excess.

Reconstruction errors, isolation-based scores, and compact learned representations encode different assumptions about unusual events. Deep SVDD learns a representation concentrated about a fixed center [3]; Deep SAD extends that principle using known anomalous examples [4]. Collider applications also include weakly supervised signal-region classifiers [5], conditional density ratios [6], and classification against generated reference events [7]. Consequently, neither limited supervision nor a combination of anomaly ranking and a resonance search is intrinsically a new method.

This work asks three narrower empirical questions. First, how do a small number of known anomalous events and unlabelled signal contamination affect ranking within one fixed feature representation? Second, how much of that ranking transfers to a different decay topology? Third, do improved rankings coincide with acceptable changes to the background mass distribution? We

report the evidence available in the released implementation, including negative outcomes, and distinguish training variability from uncertainty in the finite evaluation sample.

The principal contribution is the common protocol, released checkpoints and scores, and an implementation audit that makes the comparison reproducible. It is not a claim of state-of-the-art performance: the published density-estimation methods are not reproduced faithfully by the exploratory proxies, and a physics-motivated subjettiness cut remains an important reference. All numerical conclusions below refer to simulated samples and the stated observables, rather than detector data or an unrestricted family of possible signals.

II. DATASET AND EXPERIMENTAL PROTOCOL

A. Samples, observables, and units

The official research and development sample contains one million quantum chromodynamics background events and 100,000 nominal signal events. The nominal process is $W' \rightarrow XY$ with $X, Y \rightarrow q\bar{q}$, masses $m_{W'} = 3.5$ TeV, $m_X = 500$ GeV, and $m_Y = 100$ GeV. An additional 100,000-event sample changes the daughter decays to three-prong jets. The data were generated with Pythia and Delphes, without pileup or multiple parton interactions, and use an anti- k_T jet definition with radius one and a leading-jet trigger threshold of 1.2 TeV [1, 2]. These are properties of the supplied simulation, not generation choices newly studied here.

We use the supplied high-level jet observables rather than claim an independent reconstruction of their com-

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TABLE I. Fixed core split after high-level feature validity checks. The three-prong sample is an independent evaluation pool.

Sample	Training	Validation	Test
Background	599952	199984	199984
Two-prong signal	59983	19994	19996
Three-prong signal	0	0	99975

plete clustering and subjettness configuration. The mass-blind input vector is

$$x = (m_{J_1}, |m_{J_1} - m_{J_2}|, \tau_{21}^{J_1}, \tau_{21}^{J_2}, \Delta R_{JJ}), \quad (1)$$

where $\tau_{21} = \tau_2/\tau_1$ is a subjettness ratio [8]. The dijet invariant mass m_{JJ} is retained for diagnostics and the sideband study; it is also supplied to the mass-aware reference as a sixth feature. We use the high-energy convention $c = 1$: masses and momenta are quoted in GeV or TeV, while angular distances and subjettness ratios are dimensionless. Standardized network inputs and anomaly scores are dimensionless.

Removing nonfinite or invalid high-level rows excludes 107 events from the combined nominal sample and 25 from the three-prong sample. The retained counts are shown in Table I. The three-prong events are reserved for evaluation throughout the core comparison. This topology-transfer test shares simulation assumptions and resonance masses with the nominal hypothesis; it is not a generator-systematics or mass-extrapolation test.

B. Separation of training, validation, and test information

A class-stratified 60/20/20 split is generated with seed 42 and reused across configurations. Subsequent training seeds change initialization, ordering, and signal sampling, but do not create independent test datasets. For the autoencoder, Isolation Forest, Deep SVDD, Deep SAD, and mass-aware reference, feature means and standard deviations are fitted on training background only. The fully supervised reference instead fits its scaler on all nominal training events. Both procedures exclude validation and test events; the preprocessing difference nevertheless prevents attributing the entire supervised advantage to labels alone.

We define the contamination configuration f in percent relative to the training-background count, and inject

$$n_{\text{inj}} = \left\lfloor \frac{f}{100} N_{B,\text{train}} \right\rfloor \quad (2)$$

nominal signal events into the unlabelled training pool. Thus $f = 0.5$ means 0.5%, not 50%. Deep SAD additionally receives k known anomalies drawn without overlap with the injected unlabelled signal events. The headline

$f = 0.5$, $k = 100$ configuration contains 2,999 unlabelled signal events. Labels for the test pools never enter detector optimization.

The fully supervised network uses all 59,983 training signal labels. Its configuration name contains a $k = 1000$ field, but that field is metadata and does not restrict its training labels. We therefore describe this model as a fully supervised reference, not as a 1,000-label competitor. Supervised validation also uses both classes; the other neural models select checkpoints using a background validation objective.

C. Configuration coverage

At seed 42, the core grid contains four contamination settings $f \in \{0, 0.1, 0.5, 1\}$ for each of the three unsupervised methods, nine Deep SAD settings from $f \in \{0, 0.5, 1\}$ and $k \in \{10, 100, 1000\}$, and two reference configurations. Six headline configurations are repeated at seeds 43–46, giving 47 fits in total. Five-seed results therefore apply to the headline settings, not to every point in the contamination or label-budget grid. The canonical results table also retains five separately identified exploratory high-level configurations.

These settings were examined on a public, previously studied signal. The release tag freezes artifacts for reproducibility; it does not turn a retrospective comparison into a prospectively blinded discovery experiment. There is no statistically independent validation of a configuration chosen using its reported test AUC.

III. DETECTORS AND OPTIMIZATION

A. Unsupervised baselines

The autoencoder has widths $5 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 5$, biased linear layers, and LeakyReLU activations. Its training loss and score are the mean squared reconstruction error,

$$L_{\text{AE}} = \frac{1}{5|U|} \sum_{i \in U} \|x_i - \hat{x}_i\|_2^2, \quad s_{\text{AE}}(x) = \frac{\|x - \hat{x}\|_2^2}{5}. \quad (3)$$

An error measures incompatibility with this learned reconstruction, not a calibrated likelihood ratio. This distinction is relevant to collider autoencoder searches [9].

Isolation Forest uses 100 trees, automatic contamination and subsampling settings (256 events per tree in the employed implementation), and a seed-specific random state. Higher anomaly scores are obtained by negating the decision function. The construction is the existing isolation-based algorithm [10]; it has no neural trainable-parameter count comparable to those below.

The Deep SVDD embedding has bias-free widths $5 \rightarrow 64 \rightarrow 32 \rightarrow 16$ and ReLU hidden activations. Let $q_i = \|\phi_\theta(x_i) - c\|_2^2$. Its objective is the minibatch mean of

q_i . The center c is initialized from the mean embedding of the initial unlabelled pool and then fixed; small nonzero components are clamped to magnitude 0.1, whereas exact zeros remain zero. No autoencoder pretraining is used. In contaminated configurations the initialization pool includes the injected signal. These implementation choices matter when interpreting a near-chance outcome; that outcome alone does not establish representation collapse.

B. Limited supervision and references

Deep SAD uses the same embedding and center prescription. With unlabelled and known-anomalous subsets U_b, A_b in batch b , the implementation minimizes

$$L_b = \frac{1}{|U_b|} \sum_{i \in U_b} q_i + \frac{\eta}{|A_b|} \sum_{j \in A_b} \frac{1}{q_j + \epsilon}, \quad (4)$$

where $\eta = 1$ and $\epsilon = 10^{-6}$. The second term is omitted for a batch containing no labelled anomaly. The terms encourage small distances for the unlabelled reference and large distances for known anomalies. This is an implementation of the established Deep SAD principle [4], with separately averaged batch terms. It is not a uniformly weighted global loss over all samples. Labelled examples are mixed into the ordinary batch sampler without oversampling, so their frequency and the batch composition change the effective training dynamics as k changes. Early stopping uses background distance alone rather than Eq. (4) on a labelled validation pool.

The supervised reference has widths $5 \rightarrow 64 \rightarrow 32 \rightarrow 1$, biased layers, ReLU hidden activations, and binary cross-entropy with logits. Its sigmoid output is the anomaly score. The mass-aware reference replaces the SVDD input width by six and otherwise uses the same training prescription. This tests explicit mass inclusion in that particular hypersphere architecture, not in every anomaly detector.

Neural models use Adam with learning rate 10^{-3} , weight decay 10^{-5} , batch size 256, at most 50 epochs, and early-stopping patience eight. Parameter counts are 5,973 for the autoencoder, 2,880 for each five-feature hypersphere network, 2,497 for the supervised reference, and 2,944 for the mass-aware model. Capacities and validation objectives are not identical across methods; this is a comparison of specified practical implementations rather than an architecture-matched causal experiment.

IV. EVALUATION AND UNCERTAINTY

A. Ranking and topology transfer

For a score increasing with anomalousness, the empirical AUC estimates

$$\Pr[s(X_S) > s(X_B)] + \frac{1}{2} \Pr[s(X_S) = s(X_B)]. \quad (5)$$

The nominal test pool contains 199,984 background and 19,996 signal events, a signal prevalence of 9.0899%. This deliberate evaluation mixture is distinct from the much smaller training injection. AUC is a ranking statistic and does not imply the same precision, discovery reach, or background rate at arbitrary physical signal prevalence.

We also inspect background rejection at specified signal efficiencies and the significance-improvement characteristic $\epsilon_S/\sqrt{\epsilon_B}$ on ROC points with $\epsilon_B > 0$. This expression describes the background-dominated counting approximation; it is not a replacement for a nuisance-aware likelihood or a measure at a finite zero-background tail. ROC-selected test thresholds are descriptive only. Topology transfer is assessed using the same frozen detector score on the three-prong pool and the shared background test events.

For five-seed results we report the arithmetic mean and sample standard deviation, not a standard error or confidence interval. Separately, 500 stratified paired bootstrap resamples of the frozen seed-42 test scores use seed 20261003. Each resample preserves the class sample sizes, uses common event multiplicities across models, and handles tied scores with half credit. Percentile 95% intervals condition on the already trained detector and shared test distribution. They do not include optimization uncertainty, detector systematics, or uncertainty caused by choosing a model after inspecting the test results.

B. Mass sculpting and calibrated score cuts

Cuts intended to retain 10% or 1% of background are calibrated on validation background and frozen before test evaluation. Ties are accepted fractionally to attain the intended validation efficiency in expectation. The resulting weighted counts represent randomized acceptance probabilities; they are not necessarily integer event counts.

We quantify background mass changes through the Jensen-Shannon divergence

$$J(P, Q) = \frac{1}{2} D_{\text{KL}}(P \| M) + \frac{1}{2} D_{\text{KL}}(Q \| M), \quad (6)$$

$$M = \frac{1}{2}(P + Q).$$

using natural logarithms and normalized histograms with 49 equal-width bins between 2 and 5 TeV. Here P is the inclusive test-background mass histogram and Q is the selected histogram. We also record Pearson correlation between score and mass. Low binned divergence or low linear correlation does not establish conditional independence, nor rule out a localized feature that compromises a resonance search.

C. Approximate local-excess diagnostic

The sideband study fits 40 bins between 2.5 and 4.5 TeV, each 50 GeV wide, excluding the fixed signal win-

dow 3.3–3.7 TeV. With $x = m_{JJ}/(13000 \text{ GeV})$, the expected bin counts follow

$$\ln \lambda(x) = a + b \ln(1 - x) - c \ln x - d(\ln x)^2. \quad (7)$$

Thus logarithms have dimensionless arguments and λ denotes a count per fixed bin. Sideband Poisson fits use bounded optimization and the background yield in the window is extrapolated from Eq. (7). An inverse-Fisher approximation propagates parameter uncertainty into an estimated σ_B^2 .

For an observed window count $n > B$, the reported local diagnostic is the Gaussian-background-uncertainty approximation [11]

$$Z^2 = 2 \left[n \ln \frac{n(B + \sigma_B^2)}{B^2 + n\sigma_B^2} - \frac{B^2}{\sigma_B^2} \ln \left(1 + \frac{\sigma_B^2(n - B)}{B(B + \sigma_B^2)} \right) \right]. \quad (8)$$

We set $Z = 0$ for $n \leq B$ and use the zero-uncertainty limit $Z^2 = 2[n \ln(n/B) - (n - B)]$ when appropriate. This checks a useful analytic limit but does not validate the likelihood assumptions in the actual samples.

For these diagnostics, 999 nominal signal events are added to the 199,984 test background events, corresponding to 0.4995% relative to background or 0.4971% of the mixture. The same sampled signal subset is used before and after a cut for a given run. Crucially, nominal fits can contain fractional tie-acceptance weights, for which an ordinary Poisson count model is not exact. The nuisance approximation does not jointly profile the sideband likelihood, selection-threshold uncertainty, or background-model misspecification. Reported Z values therefore describe this diagnostic implementation and must not be interpreted as calibrated discovery significances.

V. RESULTS

A. Headline comparison

Table II summarizes the six repeated configurations. Isolation Forest is the strongest learned unsupervised core detector for the nominal topology. Deep SAD with 100 known anomalies improves its mean AUC by 0.0812, while retaining a smaller mean mass-histogram divergence at the 10% background cut. The fully supervised reference gives the highest AUC but consumes all nominal training signal labels and uses different train-only normalization and validation information. Its advantage is evidence for the value of this richer reference setup, not an isolated measurement of a label-budget effect.

At seed 42 the paired Deep SAD minus Isolation Forest difference is 0.0596 with 95% interval [0.0570, 0.0625]. Its difference from the fully supervised reference is -0.0898 , with interval $[-0.0924, -0.0874]$. Conditional finite-test errors are substantially smaller than the autoencoder’s

five-seed dispersion. Table III therefore cannot justify replacing repeated optimization runs by a single narrow bootstrap interval.

Deep SVDD ranks the nominal signal approximately at chance, with mean AUC 0.5002. This is an established observation for the specified implementation. Representation contraction or an unfavorable optimization path is a plausible explanation, but neither latent-dispersion diagnostics nor the pretraining and center-initialization ablations required to establish that mechanism are available. It would be speculative to describe the result as a general failure theorem for hypersphere anomaly detection.

A deterministic subjettness-based cut in the exploratory archive reaches nominal AUC about 0.891 and three-prong AUC about 0.676. It exceeds Isolation Forest on the nominal hypothesis, so the latter is not the best unsupervised reference overall. Such a cut is informed by the boosted-jet structure of this sample [8]; its competitiveness emphasizes the need for physics-motivated baselines when assessing the value of learned representations.

B. Contamination and known-anomaly budgets

The single-seed contamination sweep is given in Table IV. The SVDD nominal AUC decreases from 0.5123 to 0.4439 between zero and 1% contamination. Isolation Forest changes less across these four settings. These measurements do not establish monotonic population responses or causal mechanisms because most grid points lack repeated seeds and the sample of injected signal also changes.

In Table V, the 1,000-label settings give AUCs near 0.93, while the 10-label settings vary substantially. The 100-label reference is a useful intermediate operating point, rather than a statistically demonstrated optimum. Because Eq. (4) separately averages the labelled term within batches and omits it when no labels are present, a budget sweep changes both information and optimization dynamics. Controlled resampling and matched gradient exposure would be needed to separate these effects.

C. Transfer and mass dependence

The mean AUC falls by 0.1210 for the autoencoder, 0.1237 for Isolation Forest, and 0.1388 for Deep SAD when transferring from two-prong to three-prong signal. The fully supervised reference loses 0.0367. SVDD and the mass-aware variant instead increase to 0.6170 and 0.6670, respectively, from poor nominal rankings. These signed differences demonstrate topology dependence of the scores; an increase from a near-chance nominal result should not be presented as general-purpose anomaly sensitivity.

TABLE II. Five-seed headline results (seeds 42–46). AUC uncertainties are sample standard deviations across training seeds. The JSD column is a mean at the validation-calibrated 10% background cut. Deep SAD uses $f = 0.5\%$, $k = 100$; other rows use their uncontaminated reference configurations.

Detector	Two-prong AUC	Three-prong AUC	Mean mass JSD
Autoencoder	0.7175 ± 0.0741	0.5965 ± 0.1026	0.0168
Isolation Forest	0.8068 ± 0.0077	0.6831 ± 0.0206	0.0287
Deep SVDD	0.5002 ± 0.0072	0.6170 ± 0.0114	0.0012
Deep SAD	0.8880 ± 0.0114	0.7492 ± 0.0614	0.0066
Supervised	0.9659 ± 0.0002	0.9292 ± 0.0012	0.0082
Mass-aware SVDD	0.5151 ± 0.0105	0.6670 ± 0.0233	0.0262

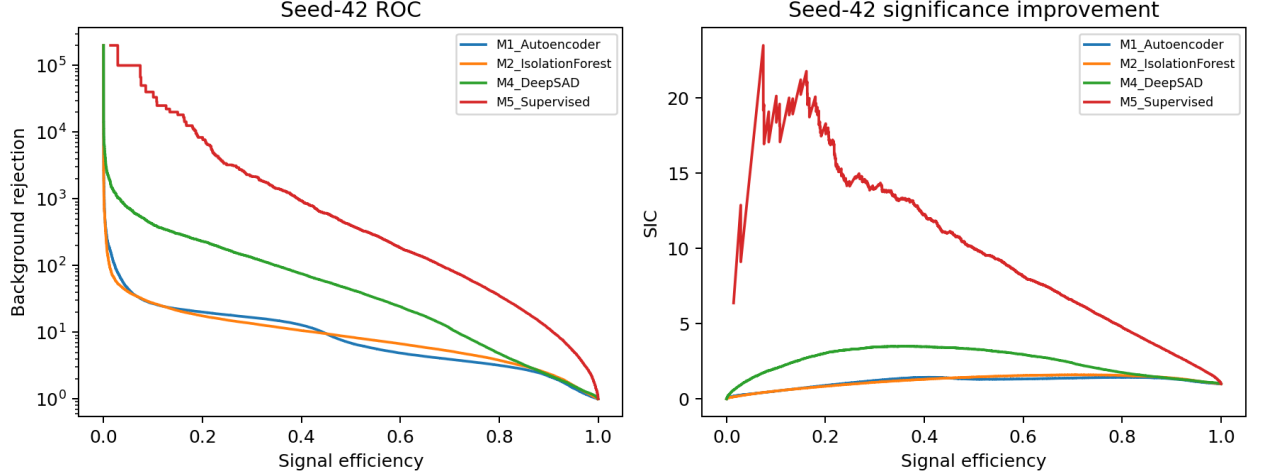


FIG. 1. Released seed-42 headline ROC curves and significance-improvement diagnostics. These single-seed curves supplement the five-seed AUC table; their counting approximation is not a discovery likelihood.

TABLE III. Seed-42 AUC and conditional finite-test 95% bootstrap interval. These intervals do not measure training-seed variation.

Detector	AUC	Interval
Autoencoder	0.7944	[0.7911, 0.7976]
Isolation Forest	0.8165	[0.8137, 0.8194]
Deep SVDD	0.5123	[0.5084, 0.5161]
Deep SAD	0.8762	[0.8735, 0.8787]
Supervised	0.9659	[0.9646, 0.9673]
Mass-aware SVDD	0.5278	[0.5239, 0.5314]

TABLE IV. Nominal AUC across the seed-42 unlabelled-contamination sweep. f is expressed in percent of training background.

f (%)	Autoencoder	Isolation Forest	SVDD
0	0.7944	0.8165	0.5123
0.1	0.7035	0.8172	0.4866
0.5	0.7160	0.8179	0.4790
1	0.7240	0.8136	0.4439

TABLE V. Seed-42 Deep SAD nominal AUC as a function of contamination and number of known anomalies. The larger budget improves this grid's ranking; no optimal budget is inferred.

f (%)	$k = 10$	$k = 100$	$k = 1000$
0	0.7611	0.8946	0.9282
0.5	0.8452	0.8762	0.9316
1	0.7695	0.8892	0.9258

Excluding m_{JJ} does not make a detector mass-independent: jet masses and substructure can be correlated with the event mass. The mean 10% mass JSD is 0.0287 for Isolation Forest, 0.0066 for the Deep SAD headline, and 0.0082 for the supervised reference. The SVDD value 0.0012 accompanies near-chance nominal discrimination. Thus a small mass-histogram change is insufficient as a success criterion. Explicit mass inclusion in the specified SVDD reference yields JSD 0.0262 and nominal AUC 0.5151, without improving the nominal task.

The sideband statistic can rise substantially after a

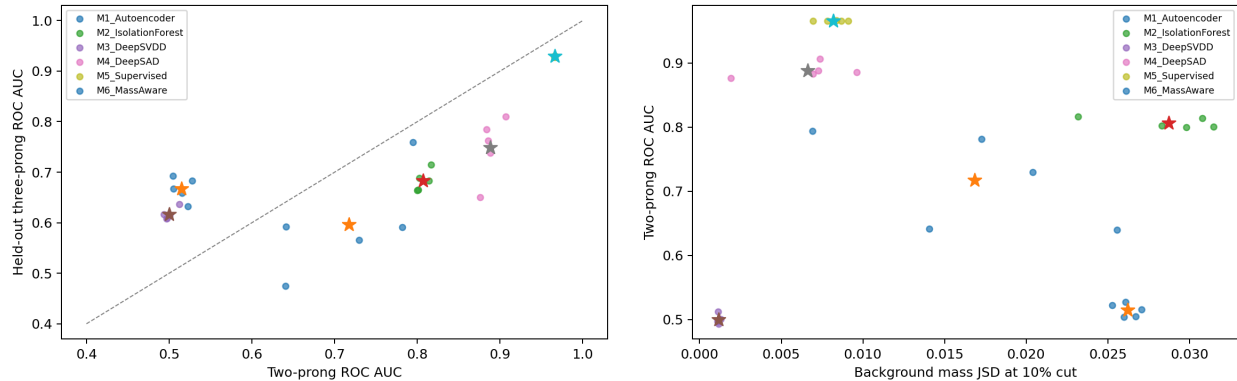


FIG. 2. Released headline topology-transfer (left) and discrimination-mass-distortion (right) diagnostics. Dots show the five training seeds and stars their means for each headline setting in Table II. The mass comparison uses the 10% background operating point. Binned JSD is a diagnostic, not a certificate of background-model adequacy.

cut. For example, at seed 42 the Deep SAD diagnostic gives $Z = 5.89$ and 7.58 at the 10% and 1% background operating points. The corresponding supervised values are 6.63 and 13.69 . Table VI reports all five reference seeds, including the failed Deep SAD 1% fit at seed 44. Omitting a failed fit from an average or a null distribution changes the estimand; these outputs cannot establish a reliable significance gain.

VI. EXPLORATORY EXTENSIONS

The following results are archived, largely single-seed extensions. They were not part of the 47-fit clean core rerun and are not combined with its seed-dispersion estimates. Their purpose is to identify follow-up questions, not to enlarge the validated claim by counting heterogeneous experiments together.

A. Particle-level reconstruction

A background-only permutation-invariant autoencoder was trained for eight epochs on a seeded subset of 50,000 background training events and selected using all 199,984 validation background events. It uses masked particle encodings and mean/max pooling of up to 700 particle triples, followed by reconstruction of valid particle features. The inputs encode normalized transverse momentum, its logarithmic transforms, pseudorapidity, and sine/cosine of azimuth. The logarithm of $1 + p_T$ in the numerical implementation assumes p_T measured in GeV; its dimensionally explicit interpretation is $\ln(1 + p_T/(1 \text{ GeV}))$. No signal training labels are used in this extension.

Nominal AUC is 0.4980 on the core test background and signal events, while the raw three-prong sample gives 0.4409 . The latter uses all 100,000 raw events, rather than the 99,975-event high-level validity-filtered pool. At

a frozen 1% validation-background threshold the nominal and three-prong signal efficiencies are 0.160% and 0.098% . On the additional public $X \rightarrow YY' \rightarrow 4q$ and $W_{KK} \rightarrow WR \rightarrow 3W$ samples [14], evaluated against a fixed 20,000-event background subset, AUCs are 0.4647 and 0.5068 ; signal efficiencies at that same threshold are 0.159% and 0.280% , respectively. The two signal pools contain 44,673 and 32,801 events and use a different generation chain from the original benchmark. These sample-specific negative results do not demonstrate absence of useful particle-level information. Architecture, reconstruction target, training duration, and transfer of the reference distribution remain untested explanations.

Semi-supervised permutation-invariant collider detectors already exist [12], and recent work studies optimal-transport event representations [13]. The present small reconstruction experiment neither reproduces those approaches nor establishes a new general limitation of constituent representations.

B. Density proxies and calibration

The archived “ANODE”-named configuration is a five-component Gaussian mixture fitted to background sidebands, with negative log background density as its score. It does not estimate the signal-region data density, form an ANODE likelihood ratio, or implement mass-conditional interpolation. Likewise the “CATHODE”-named conditional variational-autoencoder extension is a generated-reference classifier proxy, rather than the published conditional-flow construction [6, 7]. Their observed AUCs, approximately 0.720 and 0.481 , cannot be used to infer the performance of the original methods.

For the frozen mixture score, the held-out sideband pool contains 128,112 events, split equally into calibration and evaluation sets using seed 2026. The conformal

tail value is

$$p(x) = \frac{1 + \sum_{i=1}^{n_{\text{cal}}} \mathbf{1}[s_i \geq s(x)]}{n_{\text{cal}} + 1}. \quad (9)$$

Under exchangeability of calibration and future background scores and a fixed scoring model, this standard rank construction is conservative at finite sample size [15]. The measured evaluation rejection rates are 1.054%, 4.975%, and 9.868% at nominal 1%, 5%, and 10% levels. The reported uniformity diagnostic has a Kolmogorov–Smirnov statistic of 0.0043 and nominal $p = 0.179$; common calibration events and discrete ranks mean this diagnostic is not a stand-alone proof of uniform independent draws. The exercise checks same-sideband score calibration, not background-density correctness or transfer into a signal region. Recent collider work explicitly addresses conformal calibration, mass sculpting, and look-elsewhere effects [16]; Eq. (9) itself is not a novel contribution here.

VII. DISCUSSION AND LIMITATIONS

The strongest supported empirical conclusion is that a modest set of known anomalies improves the specified hypersphere detector’s nominal ranking in this benchmark, while sensitivity to a changed topology and mass distortion remain separate concerns. That conclusion is consistent with the existing purpose of Deep SAD [4]; it establishes an implementation-specific measurement, not a new learning principle. The negative SVDD and particle-level results are reproducible observations with unresolved causes.

Several limitations constrain generalization. First, the same fixed simulated test events are used across seeds and configurations. The bootstrap intervals quantify sampling uncertainty conditional on these distributions, not detector response, hadronization, or unseen resonance masses. Second, only six configurations have five-seed coverage, and no prospective hyperparameter selection or independent final holdout is claimed. Third, the supervised scaler and validation objective differ from those of the other models, while label exposure in Deep SAD changes with minibatch composition. Fourth, a physics-motivated cut and faithful modern density/particle baselines require matched-protocol comparisons before a claim of competitive novelty.

The significance diagnostic is especially limited. Bounded sideband fits require at least six positive bins, numerical convergence, and a fitted total sideband yield within a factor two of the observed yield. Fifty background-only Poisson bootstrap trials per core configuration are too few to resolve a rare discovery tail. An older 500-trial closure exercise for the autoencoder, Isolation Forest, and the conditional-generator proxy retained 461, 479, and 456 valid fits, respectively, at the 1% background cut after failures. These archived checks

concern older checkpoints, and conditioning on fit survival can distort a null distribution. Neither a small observed false-positive fraction nor a large approximate Z validates final-model discovery inference. A future study must define outcomes for failed fits, calibrate the complete procedure, and account for selection and any mass scan.

The clean core reproduction demonstrates executability, but not bitwise portability of every neural result. Historical and fresh results differ by as much as about 0.211 in autoencoder AUC and 0.171 in Deep SAD AUC at overlapping configurations, whereas some other reference scores agree. The release makes the fresh environment and outputs canonical; the causes of the neural differences have not been isolated in a factorial environment or optimization study. It would be unjustified to attribute them solely to a particular thread setting or library version.

Finally, a public benchmark with one principal family of simulated signals cannot establish a discovery on experimental data. The exploratory evaluations of additional hypotheses are too limited to replace a prospectively defined, multi-hypothesis validation campaign.

VIII. CONCLUSIONS

We provide an audited comparison of six reference detectors and a limited contamination/label-budget grid on a common LHC Olympics feature representation. Five-seed results show useful nominal discrimination for Isolation Forest, stronger ranking for a Deep SAD configuration using 100 known anomalies, and the largest AUC for a fully supervised reference. Finite-test paired bootstrapping confirms the seed-42 ranking differences within its conditional scope. A three-prong evaluation exposes topology sensitivity, while mass diagnostics show that good ranking and benign background shapes cannot be inferred from each other.

The released evidence supports an empirical technical study with transparent negative results and reproducibility artifacts. It does not establish a new anomaly-detection algorithm, a state-of-the-art comparison, or a calibrated discovery significance. The next scientific step is a focused mechanism study with matched preprocessing and strong baselines, prospectively fixed evaluation on additional hypotheses, and a statistically valid treatment of the full background-selection and sideband-fit procedure.

DATA AND CODE AVAILABILITY

The dataset is publicly available through Zenodo [2]. The code and core result artifacts are available at <https://github.com/lihtim-kitsa/lhc-olympics-2020>, release v1.0.1, commit 4b15306cfac9e3f57ace098932db0e4ea69e0f2a.

TABLE VI. Approximate local-excess diagnostic of Eq. (8) after nominal validation-background cuts. “Failed” is retained explicitly. Values are not calibrated discovery significances.

Detector	Seed	10% cut	1% cut
Deep SAD	42	5.89	7.58
Deep SAD	43	4.71	5.80
Deep SAD	44	4.13	Failed
Deep SAD	45	4.95	7.51
Deep SAD	46	4.87	8.43
Supervised	42	6.63	13.69
Supervised	43	6.66	14.71
Supervised	44	7.76	14.86
Supervised	45	6.59	13.76
Supervised	46	6.32	13.96

Canonical tables, frozen scores, bootstrap intervals, and environment specifications accompany that release. The manuscript and its data audit are subsequent documentation artifacts; the release identifier refers to the numerical core, not to a newly tagged manuscript version. Raw datasets and local checkpoints are not all embedded in Git and require the documented retrieval/reproduction steps.

ACKNOWLEDGMENTS AND AI-USE DISCLOSURE

Author and affiliation details are retained from the existing project manuscript. Funding and contribution statements must be confirmed by the responsible researcher before submission. OpenAI Codex (GPT-6; session of October 3, 2026) assisted with repository auditing, manuscript drafting, literature retrieval, interpretation checks, and document preparation. Numerical statements were checked against saved project artifacts, and cited literature was checked against primary records. This assistance does not replace independent author re-

view of the scientific argument, citations, statistical assumptions, or final text; that review remains necessary before submission. No AI system is listed as an author.

Appendix A: Local-fit outputs and reproducibility details

The shape-fit bounds are $a \in [-50, 50]$, $b, c \in [-20, 60]$, and $d \in [-20, 20]$, with a log-count regression initializer and L-BFGS-B optimization. Fisher-matrix inversion uses a pseudoinverse with tolerance 10^{-12} . These numerical choices are part of the diagnostic procedure and can affect low-count or boundary fits. A joint likelihood with robust background-model uncertainty and integer randomized event acceptance, or an explicitly calibrated weighted-data treatment, is required for a stronger inference.

The final core used an isolated Python 3.11.9 environment on Windows, with NumPy 2.3.5, pandas 2.3.3, scikit-learn 1.7.2, SciPy 1.16.3, h5py 3.15.1, and CPU PyTorch 2.13.0. The dependency lock additionally records hdf5plugin 6.0.0 and FastJet 3.5.1.5. A fresh checkout rebuilt feature/split artifacts and reran all 47 core fits using shared checksum-verified raw data, isolated output stores, and one PyTorch thread per worker. The staged run took approximately 33.3 minutes including pauses and excluding data retrieval; this is a machine-specific observation, not a universal runtime guarantee. Raw input storage was about 3.223 GB, with approximately 61.6 MB of cached features, 10.5 MB of model artifacts, and 11.5 MB of figures, excluding environments and worker duplication. Peak memory was not measured.

Thirty repository tests passed in the release verification. These include split/scaler isolation, labelled/unlabelled disjointness, threshold ties, and AUC bootstrap consistency; software tests do not establish physics validity. The accompanying data audit derives the manuscript’s numerical tables from the canonical results without fitting new models. An independent cross-machine reproduction and human scientific review remain outstanding.

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